INFSCI 1530/2160: Data Mining
Homework 1

1 Answer of the Question 1:

While mostly the solution is coordinated using the Python Script [Estimation.py], we would like to explain the solution approach considered at the code.
Data Preparation: From the given dataset, data was filtered out and copied to an object based on the prospective season duration. Since winning threshold was 0.5, we consider binary outcome (win(1), loss(0)) and there was some tie value. Finally, our data object was generated by filtering them out.
Generated Results based on script:
Total number of games from 2010 to 2020: 2930
Accuracy of predictions from 2010 to 2020: 0.6457
Brier Score from 2010 to 2020: 0.2196
Uncertainty from 2010 to 2020: 0.2458
Resolution from 2010 to 2020: 0.0262
Reliability from 2010 to 2020: 0.0009
Decomposed Brier Score from 2010 to 2020: 0.2205

Accuracy: To calculate accuracy, at first we have filtered the dataset based on the season range provided and then find the actual outcome by finding the difference between scores and assigning probability threshold 0.5 as per question statement. After that, calculating the ratio between correct prediction (elo_prob1 vs actual outcome) and the total game count is considered as Accuracy.

Brier Score: Meanwhile the mean squared difference between actual and predicted probability has been considered as Brier Score.

Uncertainty: On the other hand, uncertainty was calculated by,

$$Uncertainty = BaseRate \times (1 - BaseRate)$$

$$BaseRate = actual_win_count/total_games$$

Resolution: For the resolution and reliability, we have considered 10 bins starting from the 0.0 to 1.0 elo_prob1 with interval 0.1. Finding the bin base_rate of each of the interval and getting the mean of the squared differences of those bins was my approach to calculate the Resolution.

$$\text{Resolution} = \sum_{k=1}^{10} \frac{(\text{Base rate of bin}_k - \text{Base rate})^2 \times \text{bin length}}{\text{total games}}$$

Reliability: In the similar way, using the formula for reliability showed us the value of it. We know that,

$$\text{Brier Score} = \text{Uncertainty} + \text{Reliability} - \text{Resolution}$$

Therefore, Brier Score found by 3 decomposed term is = 0.2458 + 0.0009 - 0.0262 = 0.2205 while the overall Brier Score was 0.2196 There is some difference found between overall brier score and decomposed brier score and it might be due to the bin preferences. Changing the number of bins might put difference reliability and resolution score.

The reliability curve graph has been also generated by the same script.

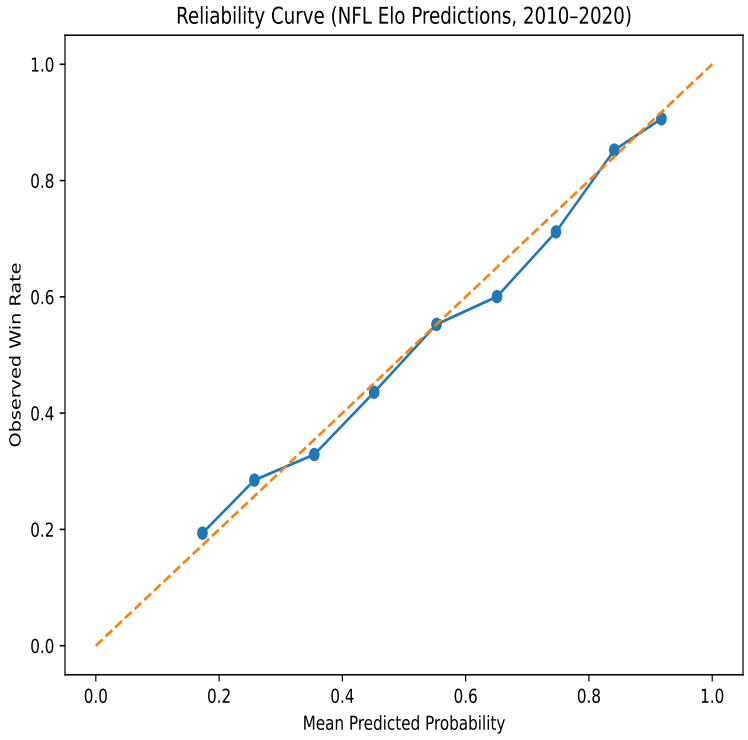


Figure 1: Reliability curve for the NFL predictions.

2 Answer of the Question 2:

The following question if any of the two version is associated with the higher sale conversion can be answered by Hypothesis testing. Let's assume that,

Null Hypothesis ($\mathbf{H_0}$): A and B website versions have the same conversion rate that is ($C_A = C_B$)

Alternate Hypothesis ($\mathbf{H_1}$): A and B websites have different conversion rate that is ($C_A \neq C_B$)

Now,

Average conversion rate of Version A, $C_A = \frac{1320}{3250}$

Average conversion rate of Version B, $C_B = \frac{1250}{2870}$

Therefore, conversion rate difference, $D_{B,A} = \frac{1250}{2870}$ - $\frac{1320}{3250} = 0.029386 = 2.94\%$. Therefore, it seems that website version B has the higher conversion rate. Now we want to check if this happens for random sampling.

We consider the pooled mean of C_A and $C_B = \frac{1320+1250}{3250+2870} = 0.4199$

Therefore, under the null hypothesis, we assume each of the version has the conversion rate is approximately 0.4199. Now, we consider having binomial distribution for our simulation with $n = 500000$ trials.

Using the simulation found by the script [HypothesisTesting.py], we have found
Observed Difference: 0.0294
Simulated p-value: 0.0197

We also observed that conversion rate B is higher than conversion rate of A. Since, simulated p-value $0.0197 \leq 0.05$, we reject the null hypothesis $\mathbf{H_0}$.Therefore, simulation also provides strong evidence that the conversion rates of A and B are different.

3 Answer of the Question 3:

Assume that,
Paying out event is, E

Given that,
Probability of choosing M_1 or $M_2 = P(M_1) = P(M_2) = \frac{1}{2} = 0.5$
Probability of M_1 machine is paying out $= P(E|M_1) = \frac{15}{100} = 0.15$
Therefore, probability of M_1 not paying out $= P(\sim E|M_1) = 1 - .15 = 0.85$
Probability of M_2 machine is paying out $= P(E|M_2) = \frac{5}{100} = 0.05$
Therefore, probability of M_2 not paying out $= P(\sim E|M_2) = 1 - .05 = 0.95$
We have to find out,

- What is the probability of being machine M_1 given that the machine does not pay out $= P(M_1| \sim E) = ?$

  We express,

$$\begin{aligned} P(M_1 \mid \sim E) &= \frac{P(\sim E \mid M_1) * P(M_1)}{P(\sim E)} \\ &= \frac{P(\sim E \mid M_1) * P(M_1)}{P(\sim E \mid M_1)P(M_1) + P(\sim E \mid M_2)P(M_2)} \\ &= \frac{0.85 * 0.5}{0.85 * 0.5 + 0.95 * 0.5} \\ &= 0.472 \end{aligned}$$

  Answer: 0.472

- What would be the next estimation given that it had pay out $= P(M_1|E) = ?$

  We express,

$$\begin{aligned} P(M_1 \mid E) &= \frac{P(E \mid M_1) * P(M_1)}{P(E)} \\ &= \frac{P(E \mid M_1) * P(M_1)}{P(E \mid M_1)P(M_1) + P(E \mid M_2)P(M_2)} \\ &= \frac{0.15 * 0.5}{0.15 * 0.5 + 0.05 * 0.5} \\ &= 0.75 \end{aligned}$$

  Answer: 0.75